Maitrayee Keskar

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Education

University of California, Merced Aug. 2026 – May 2030
Ph.D. in Electrical Engineering and Computer Science (EECS) *Advisor: Ross Greer*

University of California, San Diego
M.S. in ECE (Intelligent Systems, Robotics, Control) GPA: 3.957 | Sep. 2024 – March 2026
B.S. in Probability and Statistics (*Minor in CS, Cum Laude*) GPA: 3.896 | Sep. 2019 – June 2022

Research Experience

LISA Lab, UC San Diego Apr. 2021 – Present
Graduate & Undergraduate Student Researcher Advisor: Prof. Mohan Trivedi

- Graduate Research (with MI3 Lab, UC Merced; Advisor: Prof. Ross Greer)
  - Led the design and implementation of **MTR-VP**, a novel, vision-first trajectory planning architecture that replaces high-definition map features with learned visual representations.
  - Innovated a cross-attention decoder fusing vehicle intent with visual context from a **ViT** to generate multiple, diverse future trajectories, boosting down-stream planning performance.
  - Achieved a **top-15 rank** in the official Waymo E2E Challenge and first-authored a paper submitted to *IEEE Robotics and Automation Letters*.
- Undergraduate Research
  - Adapted the anchor-free **CenterNet** model for joint vehicle and substructure detection, leading to a first-author publication in *IEEE Robotics and Automation Letters*.
  - Engineered a custom keypoint detection head and loss function in **PyTorch**, achieving a 78.2% AP on the ApolloCar3D dataset.

MINDS Lab, UC San Diego Jan. 2025 – Jan. 2026
Graduate Student Researcher Advisor: Prof. Parinaz Naghizadeh

- Pioneered a theoretical framework to formally connect communication protocols to the sample complexity of multi-agent reinforcement learning (MARL) algorithms.
- Adapted the **Optimized Maximum Likelihood Estimation (OMLE)** algorithm to mathematically quantify the quality of an agent's observation function.
- Formulated proofs establishing formal guarantees on how specific communication schemes affect decentralized training efficiency and equilibrium convergence.

Existential Robotics Lab (ERL), UC San Diego Nov. 2024 – Sep. 2025
Graduate Student Researcher Advisor: Prof. Nikolay Atanasov

- Built an experimental validation framework for a lightweight attention-based MARL policy, contributing to a co-authored **ICRA** submission.
- Implemented a Graph Attention Network (**GAT**) baseline policy within the **BenchMARL** framework to provide a rigorous benchmark for comparison.
- Engineered a head-to-head evaluation suite from scratch, developing tracking metrics such as capture rate and inter-agent distance.

Industry Experience

Balbix San Jose, CA
Software Engineer, AI Jul. 2022 – Aug. 2024

- Designed production data pipelines using **PySpark** to analyze 500k+ enterprise vulnerability instances, offering customer-specific risk remediation paths.
- Modeled risk vectors for 2M+ End-of-Life (EOL) software packages, deploying scheduled aggregation workloads via **Airflow** and **Kubernetes**.
- Developed an event generator tracking over 5M active state changes in infrastructure asset software utilizing **Cassandra**.
- Built statistical/probabilistic models based on CVSS data, exploit kit availability, and CWE to predict breach likelihoods; delivered technical presentations directly to the CTO and VP.

Balbix San Jose, CA
Engineering Intern Jun. 2021 – Sep. 2021

- Implemented Reinforcement Learning architectures (**Q-Learning, Actor-Critic**) to model and predict adversarial lateral movement paths for simulated network defense systems.

Publications & Presentations

- Martinez-Sanchez, A., Ng, K., Maia, W., Fleig, L., **Keskar, M.**, Maquiling, E., Tandon, Y., Roy, P., Trivedi, M., Greer, R. (2026). Vision-Language Work Zone Intelligence for Safety-Critical Speed Regulation of Mixed-Autonomy Vehicles in Dynamic Environments. *IEEE International Conference on Intelligent Transportation Systems (ITSC)*. Accepted.
- Greer, R., **Keskar, M.**, Martinez-Sanchez, A., Roy, P., Shriram, S., Trivedi, M. (2026). Vision and Language: Novel Representations and Artificial intelligence for Driving Scene Safety Assessment and Autonomous Vehicle Planning. *Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV)*. National Highway Traffic Safety Administration (NHTSA). [pdf]
- Greer, R., Fleig, L., **Keskar, M.**, Maquiling, E., Tapia Lopez, G., Martinez-Sanchez, A., Roy, P., Rattigan, J., Sur, M., Vidrio, M., Marcotte, T., Trivedi, M. (2026). Looking and Listening Inside and Outside: Multimodal Artificial Intelligence Systems for Driver Safety Assessment and Intelligent Vehicle Decision-Making. *Proceedings of the 28th ESV Conference (NHTSA)*. [pdf]
- Gopalkrishnan, A., Greer, R., **Keskar, M.**, Trivedi, M. (2026). Robust Detection, Association, and Localization of Vehicle Lights: A Context-Based Cascaded CNN Approach and Evaluations. *Proceedings of the 28th ESV Conference (NHTSA)*. [pdf]
- Sebastián, E., **Keskar, M.**, Iqbal, E., Montijano, E., Sagüés, C., & Atanasov, N. (2025). Policy Gradient with Self-Attention for Model-Free Distributed Nonlinear Multi-Agent Games. *arXiv preprint arXiv:2509.18371*. (Submitted to IROS 2026). [pdf]
- **Keskar, M.**, Greer, R., Gopalkrishnan, A., Deo, N., & Trivedi, M. (2025). Lights as points: Learning to look at vehicle substructures with anchor-free object detection. *IEEE Robotics and Automation Letters*. Presented at IEEE CASE 2025. [pdf]
- Greer, R., Gopalkrishnan, A., **Keskar, M.**, & Trivedi, M. M. (2024). Patterns of vehicle lights: Addressing complexities of camera-based vehicle light datasets and metrics. *Pattern Recognition Letters*, 178, 209-215. [pdf]
- **Maitrayee Keskar**, Nachiket Deo, Ross Greer, and Mohan M. Trivedi. "A Center-Based Integrated Vehicle Internal and Landmarks Detector." Presentation at *IEEE IV ITSIVUE Workshop*, 2022. [presentation]

Technical Skills

Languages: Python, SQL, Java, C++, R, C, Shell Scripting
ML/AI Frameworks: PyTorch, TorchRL, TensorFlow, Scikit-Learn, OpenCV, Pandas, BenchMARL
Tools & Infrastructure: Kubernetes, Docker, Airflow, PySpark, Git, AWS, Postgres, Redis, Cassandra